WNDRVR
SUPPORT NETWORK

VXWORKS 7 CREATING A WEB SERVER TUTORIAL, SR0650

Copyright Notice

Copyright © 2022 Wind River Systems, Inc.

All rights reserved. No part of this publication may be reproduced or transmitted in any form or by any means without the prior written permission of Wind River Systems, Inc.

Wind River, Simics, Tornado, and VxWorks are registered trademarks of Wind River Systems, Inc. Helix, Pulsar, Rocket, Titanium Cloud, Titanium Control, Titanium Core, Titanium Edge, Titanium Edge SX, Titanium Server, and the Wind River logo are trademarks of Wind River Systems, Inc. Any third-party trademarks referenced are the property of their respective owners. For further information regarding Wind River trademarks, please see:

www.windriver.com/company/terms/trademark.html

This product may include software licensed to Wind River by third parties. Relevant notices (if any) are provided for your product on the Wind River download and installation portals:

https://delivers.windriver.com/

https://windshare.usa.windriver.com/

Wind River may refer to third-party documentation by listing publications or providing links to third-party websites for informational purposes. Wind River accepts no responsibility for the information provided in such third-party documentation.

Corporate Headquarters

Wind River
500 Wind River Way
Alameda, CA 94501-1153
U.S.A.
Toll free (U.S.A.): +1-800-545-WIND
Telephone: +1-510-748-4100
Facsimile: +1-510-749-2010

For additional contact information, see the Wind River website:

www.windriver.com

For information on how to contact Customer Support, see:

www.windriver.com/support

VxWorks 7 Creating a Web Server Tutorial, SR0650

10 June 2020

1. WEB SERVER CREATION OVERVIEW AND WORKFLOW

Learn to create a web server and demonstrate some basic functionality implemented with demonstration HTML pages.

This tutorial assumes that you have installed VxWorks® and Workbench on your host system; that you are familiar with both VxWorks and Workbench. You should also have a basic familiarity with VxWorks 7 CLI, Web, MIBway. For more information, see:

- *VxWorks 7 CLI, Web, MIBway Programmer's Guide:* Overview
- *VxWorks 7 CLI, Web, MIBway Programmer's Guide:* Fundamentals

This tutorial does not make use of the management configuration editor (MCE).

For additional tutorials, see the VxWorks 7 Creating a CLI Tool Tutorial and the VxWorks 7 Creating a CLI Tool and Web Server Tutorial.

The files for the demonstration are located in

installDir **/vxworks-7/pkgs_v2/app/webcli/webclidemo-**x.x.x.x**/**

In general, you should not edit these files beyond configuring them with MCE.

Workflow

Perform the following workflow to create the web server.

> Note: This feature only works in kernel mode for VxWorks. It is not supported for RTP projects.

1. Create and configure a web server VxWorks source build (VSB) project for the VxWorks Simulator (VxSim).
   See Creating a VSB for the Web Server on page 2.
2. Create and configure a Web server VxWorks image project (VIP) for VxSim.
   See Creating a VIP for the Web Server on page 3.
3. Create a ROM file system (ROMFS) project with a default certification file.
   The ROMFS project enables secure access to the web server using HTTP over SSL (HTTPS).
   See Creating a ROMFS Project on page 4.
4. Install the Wind River WRTAP network driver and configure network connectivity to enable IP communications between VxSim and your host.
   See Installing the Wind River TAP Driver on page 5.
5. Create a target connection between the simulator and your host.
   See Configuring a Network Connection to the Simulator on page 6.
6. Launch the VxSim daemon, vxsimnetd.
   See Starting the VxSim Network Daemon on page 8.
7. Launch the simulator instance and connect to the instance.
   See Launching the Simulator Instance and Connecting to the Simulator on page 9.
8. Access the web pages from a web browser.
   See Accessing the Demonstration Web Pages on page 10.

WNDRVR

2. CREATING A VSB FOR THE WEB SERVER

You can create a VxWorks source build (VSB) project to support a web server running on a VxSim simulator on a Windows host.

Procedure

1. Base the VSB on the VxWorks simulator by selecting either SIMNT for a CPU build or vxsim_windows_*x*_*x*_*x*_*x* for a BSP build.
2. Enable the OPENSSL, WEBCLI and WEBCLI_WEBDEMO options.
3. Verify that the SNMP option is enabled.
4. View the default configuration of the WEBCLI_COMMON parameters.

   You should retain the default configuration for the tutorial.

5. View the default configuration of the WEBCLI_HTTP parameters.

   You should retain the default configuration for the tutorial.

6. Build the VSB.

WNDRVR

3. CREATING A VIP FOR THE WEB SERVER

You can create a VxWorks image project (VIP) to support a Web server running on a VxSim simulator on a Windows host.

Procedure

1. Base the VIP on the VSB created in Creating a VSB for the Web Server on page 2.
2. Add the **INCLUDE_WEBCLI_WEBDEMO** component.
3. Verify that **INCLUDE_IPWRAP_GETHOSTBYNAME** is included in the project.
4. Build the VIP.

WNDRVR

4. CREATING A ROMFS PROJECT

You can use a read-only memory file system (ROMFS) project to include the **server.pem** file in your VxWorks image project (VIP).

About This Task

This file is automatically loaded into the simulator at boot time. For this tutorial, the **server.pem** file is loaded in the **/romfs/htdocs/** folder.

To create a ROMFS and store the **server.pem** file in the **/romfs/htdocs/** directory of the VxWorks image, proceed as follows:

Procedure

1. From the Workbench Project Explorer, select the Web Server VIP project created in Creating a VIP for the Web Server on page 3.
2. From the Workbench menu, select **File > New > Wind River Workbench Project**.
3. From the New Wind River Workbench Project dialog, set the **Build** type to ROMFS File System and name the project **My_romfs**.
4. Click **Finish** to create the ROMFS project and display the **My_romfs** project tab.
5. From the **My_romfs** project tab, click **New Folder** and create a **/htdocs** folder.

   The folder appears in the Target contents panel.

6. From the **My_romfs** project tab, click **Add External** and include the **server.pem** file located at *installDir* **/vxworks-7/pkgs_v2/app/webcli-***x.x.x.x* **/webdemo-***x.x.x.x* **/src/htdocs**.

   The **server.pem** file icon appears in the Target contents panel under the **htdocs** folder.

7. Save the ROMFS project.

   The ROMFS project files are located in your Workbench workspace under the **My_romfs** folder.

5. INSTALLING THE WIND RIVER TAP DRIVER

To simulate network IP connectivity, VxSim includes a Wind River TAP (WRTAP) driver that allows VxWorks to communicate by handing off IP packets to the host operating system for processing.

About This Task

Once installed, the WRTAP driver is automatically initialized upon launching VxSim.

For more information on using the WRTAP driver, see the VxWorks 7 VxSim User's Guide.

To install the Wind River WRTAP driver, proceed as follows:

Procedure

1. Open a Windows command shell as an administrator.
2. In the command shell, use the following command, where *installDir* is the name of your VxWorks installation directory:

   ```
   C:\> installDir\vxWorks-7\host\x86-win32\bin\wrtapInstaller.exe install wrtap
   ```

   Observe the presence of a new network connection on your host. For example, **VMware Virtual Ethernet Adapter for VMnet1**.

6. CONFIGURING A NETWORK CONNECTION TO THE SIMULATOR

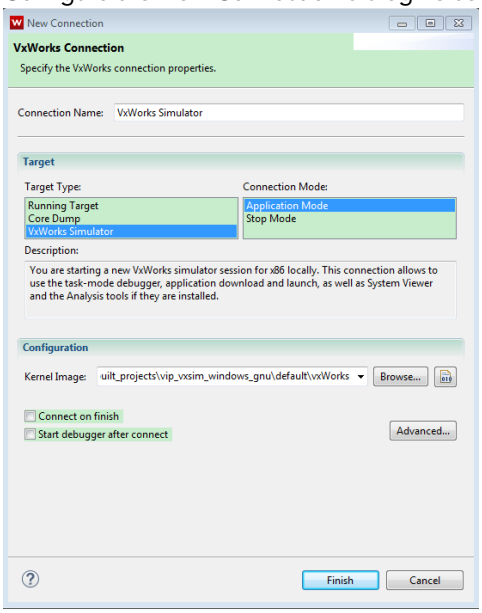
Before you can use the simulator, you must configure a network connection to it.

About This Task

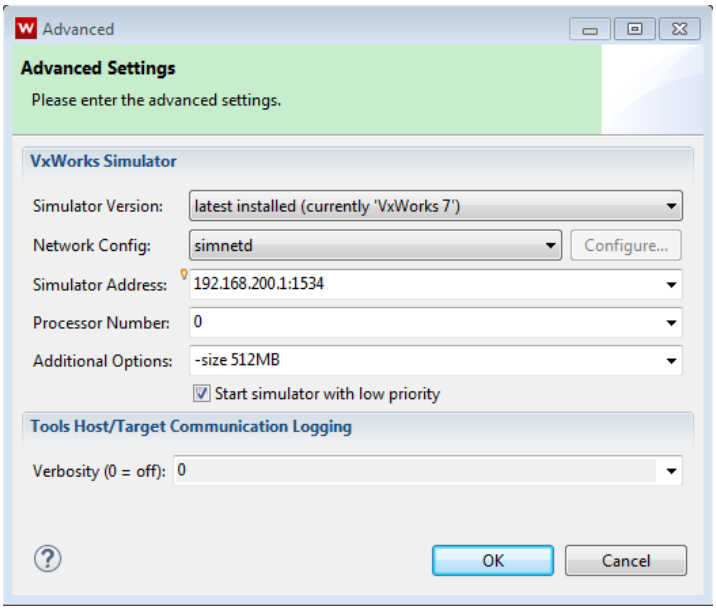
To configure a network connection between the simulator instance and your host, proceed as follows:

Procedure

1. From Workbench, select **New Connection**
2. Configure the New Connection dialog fields as follows:



3. Click **Configure** to display the Configure dialog.
4. Configure the fields as follows:



5. Click **Browse** and navigate to your VxWorks image project (VIP) default folder and select the **vxWorks** executable.
6. Click **OK**.
7. From the New Connection dialog, click **Finish**.

7. STARTING THE VXSIM NETWORK DAEMON

If you use network services, you cannot start a VxWorks simulator instance until the VxWorks simulator network daemon is started and has completed its initialization.

About This Task

For more information on the VxSim network daemon, see the VxWorks 7 VxSim User's Guide.

To start the vxsimnetd daemon with the default configuration, proceed as follows:

Procedure

1. Open a Windows command shell as an administrator.
2. In the command shell, use the following command syntax, where *installDir* is the name of your VxWorks installation directory:

```
C:\> installDir\vxworks-7\host\x86-win32\bin\vxsimnetd
```

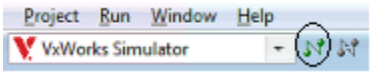

WNDRVR

8. LAUNCHING THE SIMULATOR INSTANCE AND CONNECTING TO THE SIMULATOR

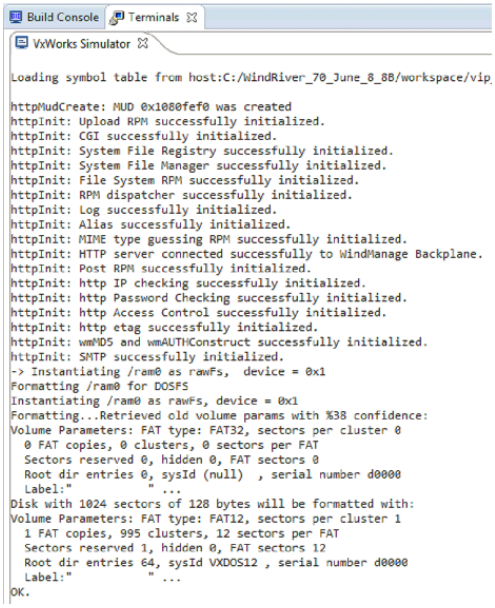
After configuring network connectivity, you can launch the simulator instance and connect to the simulator.

Procedure

1. If not started, start the VxSim network daemon. See Starting the VxSim Network Daemon on page 8.
2. Select the **VxWorks Simulator** connection from the **Connection** drop-down list and click the **Connect** icon.



You can observe the terminal output for the VxWorks Simulator. For example:



3. From the VxWorks Simulator terminal, run the ifconfig command.

   Observe the presence of the 192.168.200.1 network interface. For example:

```
-> ifconfig
...
simnet0 Link type:Ethernet HWaddr 7a:7a:c0:a8:c8:01 Queue:none
inet 192.168.200.1 mask 255.255.255.0 broadcast 192.168.200.255
inet6 unicast fe80::787a:c0ff:fea8:c801%simnet0 prefixlen 64
...
```

9. ACCESSING THE DEMONSTRATION WEB PAGES

After launching your kernel image and starting the demonstration, you can access the demonstration web pages.

Procedure

1. Launch a web browser.
2. Enter 192.168.200.1 in the address field of the web browser.

   Alternately, you can access the pages using https by entering https://192.168.200.1 in the address field of the web browser.

   You are now connected to your embedded HTTP server.

   The tutorial index page, **index.html**, appears in your browser.